

AIN SHAMS UNIVERSITY

FACULTY OF ENGINEERING

MECHATRONICS ENGINEERING DEPARTMENT



Spring 2026

MCT344 / MCT342 Industrial Robotics

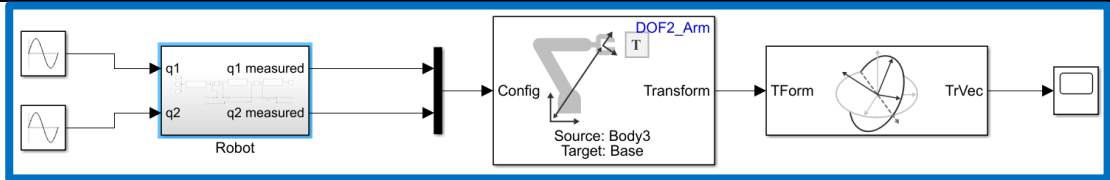
Lab Sheet 2

MATLAB & Simulink **Lab2: Robot Inverse Kinematics**

In this lab, you are required to perform the following requirements on the RR Robot you created in Lab1:

- 1- Generate a simple workspace point cloud by sampling joint configurations.
- 2- Solve inverse kinematics for target end-effector positions/poses with different initial guesses and weightings.
- 3- Verify solutions using forward kinematics and pose error metrics.
- 4- Generate a Cartesian end-effector trajectory (straight-line/interpolated poses), solve IK along the path.

Part 1: Generating a Simple Robot Workspace using MATLAB

Step	Action	Screenshot
1	We will be starting this lab with the final model of lab 1 as shown:	
	A	
	B	Run the MATLAB Script we created in lab1 that creates RigidBodyTree .
	C	MATLAB Home Tab > New > Script
	D	Copy the script below, then replace each (DOF2_Arm) with your RigidBodyTree name (5 instances)

Creating a MATLAB (.m) file

```

clc; clear points;

DOF2_Arm.DataFormat = "column"; % important
N = 15000; % number of samples
points = zeros(N,3);

eeName = DOF2_Arm.BodyNames{end}; % usually last body is end-effector

for i = 1:N
    config = randomConfiguration(DOF2_Arm); % respects joint limits
    T = getTransform(DOF2_Arm,config,eeName);
    points(i,:) = T(1:3,4)';
end

figure;
scatter3(points(:,1),points(:,2),points(:,3),5,'filled');
axis equal;
grid on;
xlabel('X'); ylabel('Y'); zlabel('Z');
title('Workspace Point Cloud');
view(2); % for planar RR robot

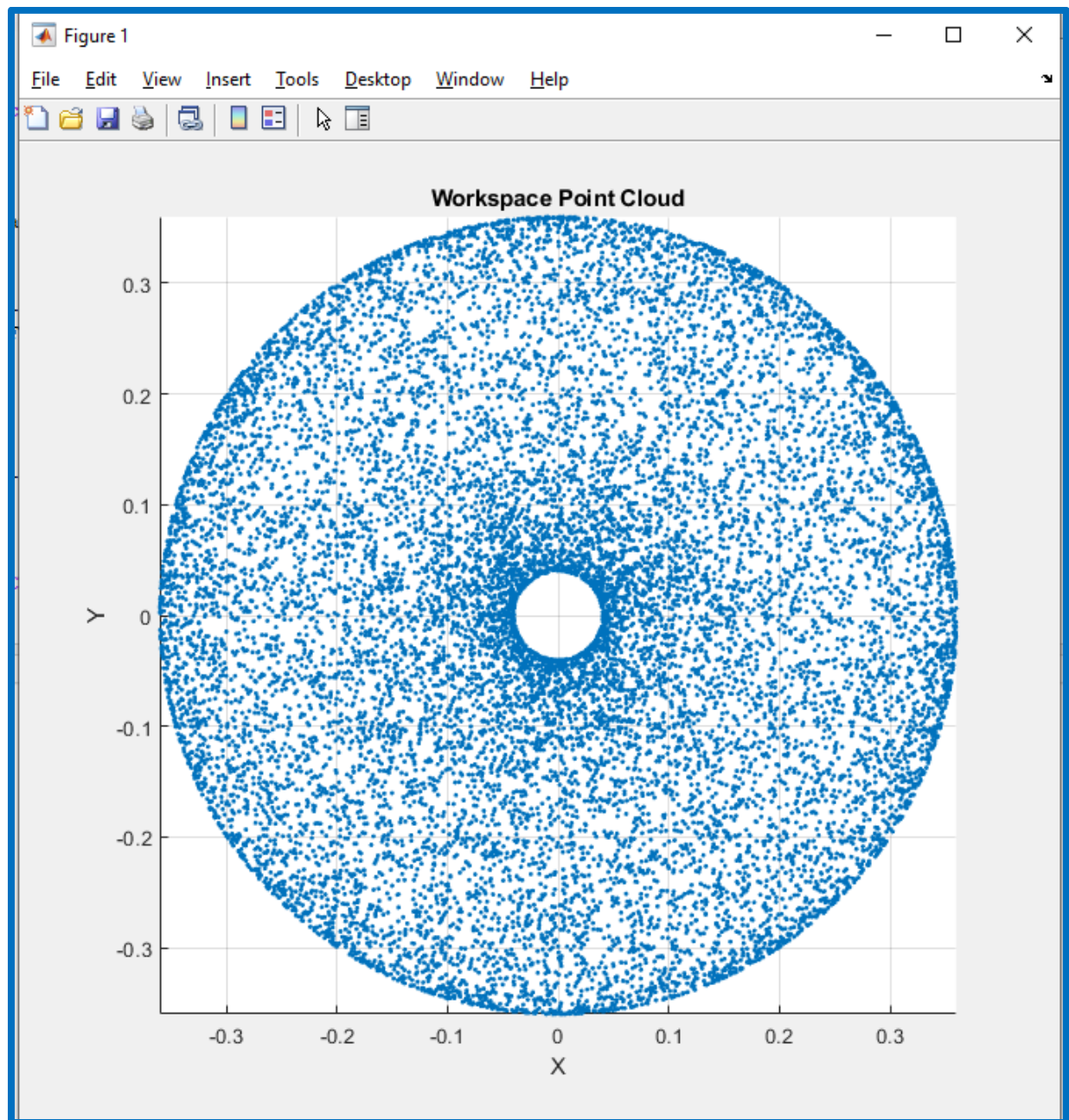
figure
scatter3(points(:,1),points(:,2),points(:,3),5,'filled')

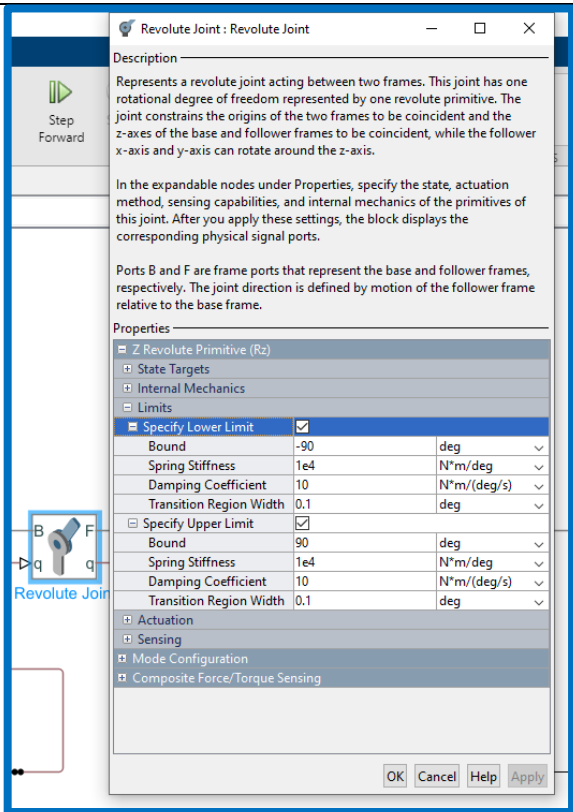
axis equal
grid on
xlabel('X')
ylabel('Y')
zlabel('Z')

title('Robot Workspace (3D)')
view(3)

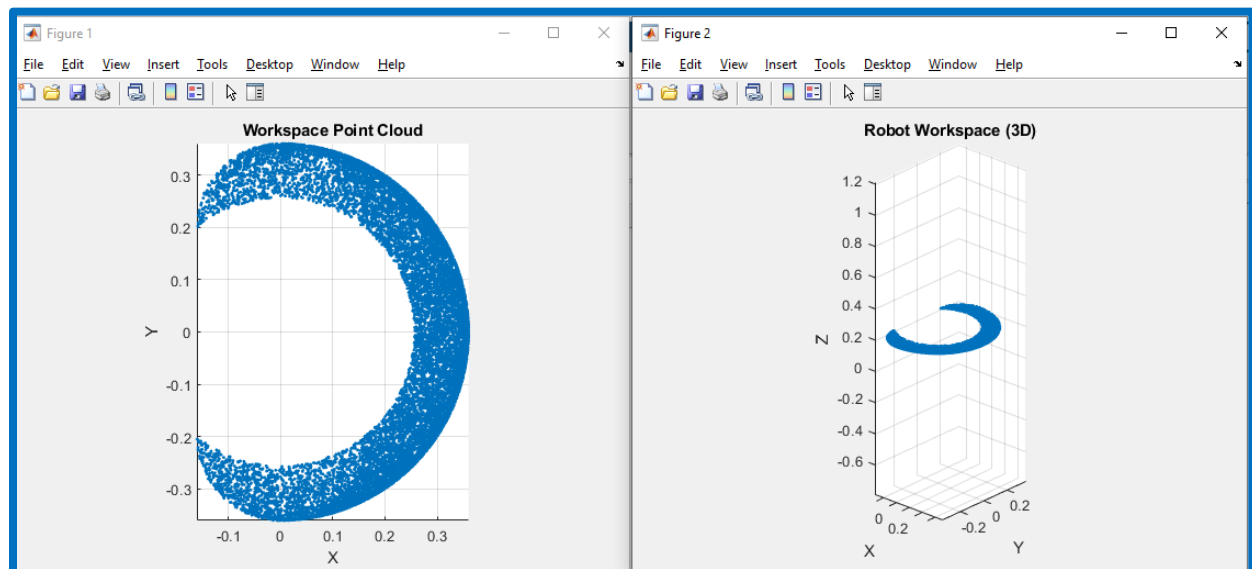
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This should appear:

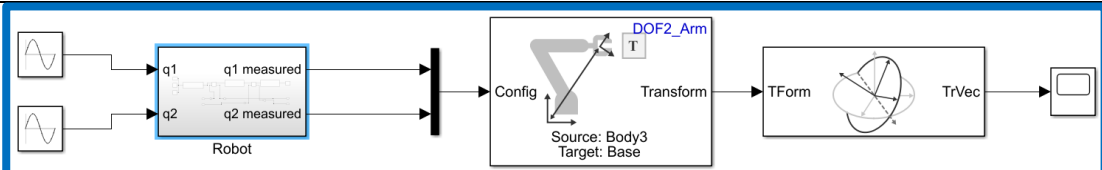
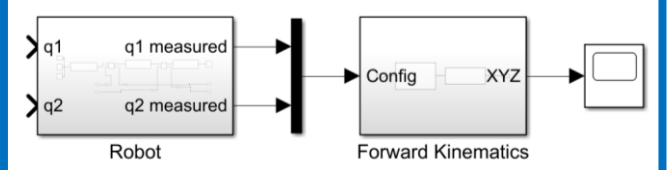


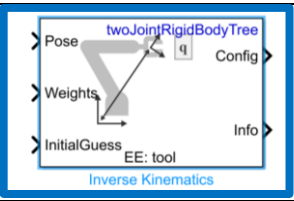
Step	Action	Screenshot
2	A Now go to the Robot Subsystem	
	B Go to Revolute Joint	
	C Limits >> Specify Lower Limit >> Check	
	Limits >> Specify Upper Limit >> Check	
	D Without changing the default values, this should limit this joint movement to 180 degrees from -90 to +90	
E	Repeat for the other revolute joint	

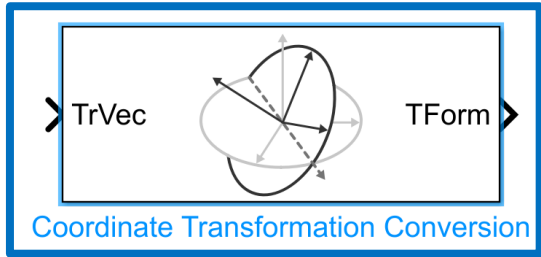
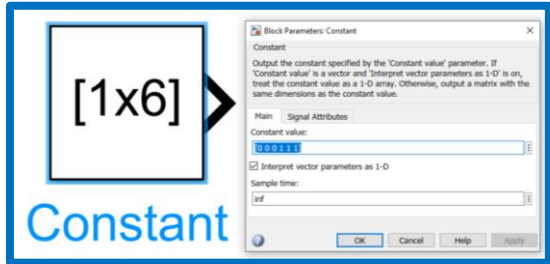
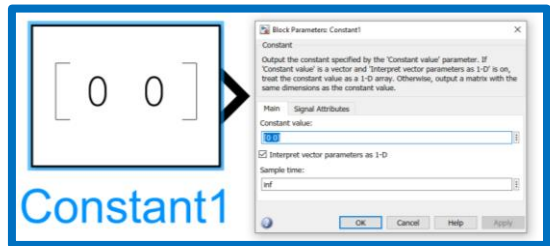
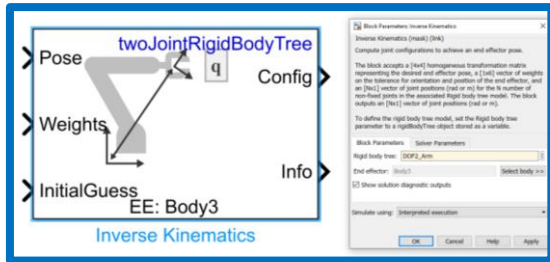
This should appear:



Part 2: Performing Inverse Kinematics on the Robot we made from Lab 1

Step	Action	Screenshot
1	A	We will be starting this lab with the final model of lab 1 as shown: 
	B	Make the “Get Transform” and “Coordinate Transformation Conversion” blocks one subsystem named: “Forward Kinematics” 
	C	Delete the Sine Wave inputs

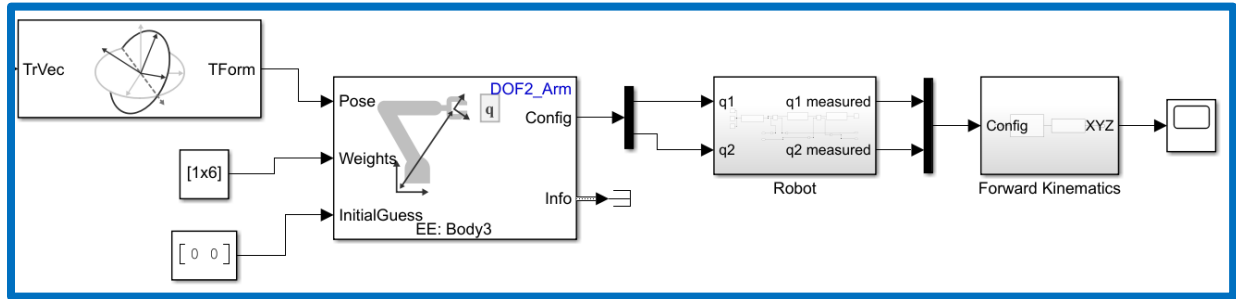
Step	Action	Screenshot
2	A	Robotic Systems Toolbox > Manipulator Algorithms > Inverse Kinematics Block 
	B	Note: Now the inverse kinematics block needs 5 inputs: 1- Pose 2- Weights 3- Initial Guess 4- Rigid Body Tree 5- End Effector

Step	Action	Screenshot
3	1- Pose: This is the 3 translation and 3 rotation variables of the end-effector. Robotic Systems Toolbox > Utilities > Coordinate Transformation Conversion Block Coordinate Transformation Conversion Block: Input >> Translation Vector Output >> Homogenous Transformation	
	2- Weights: This is a 1x6 vector that is responsible for deciding the priority when trying to solve IK. Note: the 1st 3 variables are Rotation and the 2nd 3 variables are Translation Simulink > Sources > Constant Block Constant Block: Value >> [0 0 0 1 1 1]	
	3- Initial Guess: This is a N-DOF vector that is responsible for specifying the initial guess of each joint variable for the IK block to start solving from. Simulink > Sources > Constant Block Constant Block: Value >> [0 0]	
	4- Rigid Body Tree: Inverse Kinematics Block: RigidBodyTree >> DOF2_Arm (the same as we did in get transform block in lab1)	
	5- End Effector: Inverse Kinematics Block: End Effector >> Body3 (the last body, the red Cube)	

Step	Action	Screenshot
4	A Connections: Coordinate Transformation Conversion == Pose 1x6 constant block == weights 1x2 constant block == pose	
	B - Insert a Demux block - Connect configuration of IK block to the Demux to be able to separate each joint variable. - Connect Demux to q1 and q2 of the robot	
	C - Insert a Terminator block - Connect it to Info	

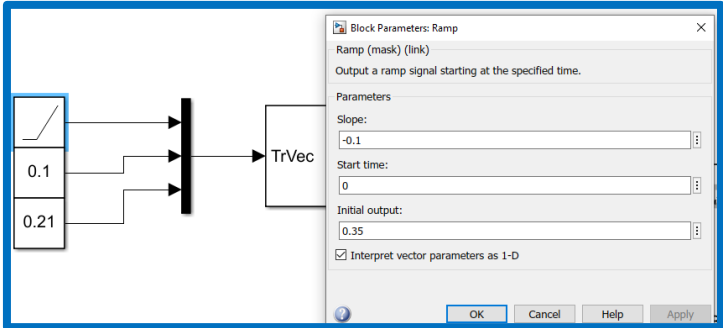
Step	Action	Screenshot
5	A Inverse Kinematics Block: Solver Parameters Tab: We will reduce the tolerances to get faster results. You can modify these values later to get more accurate results. - Solver >> Levenberg-Marquardt - Gradient Tolerance ~ 1e-4 - Solution Tolerance ~ 1e-4 - Error Change Tolerance ~ 1e-4 Note: you can make the tolerances bigger to get faster simulation	
	B Stop Time >> 5 seconds	

Your Model Should Look like this now:



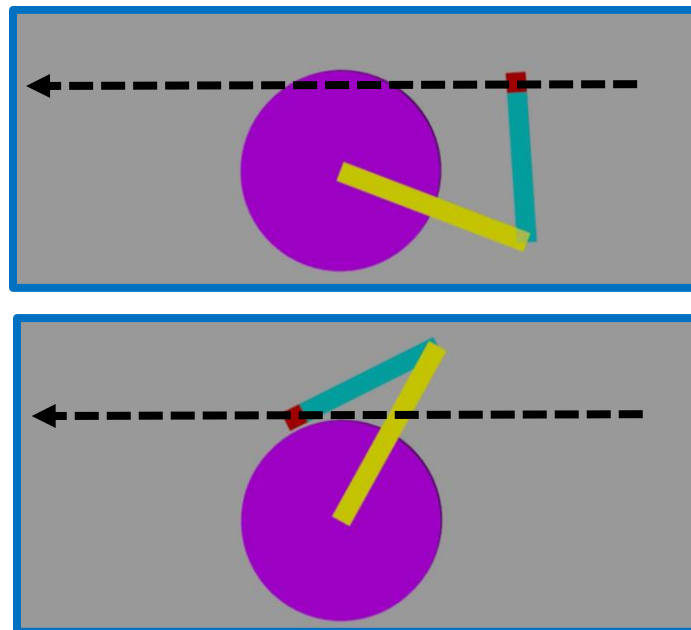
Part 3: Trying Multiple Manually-Made trajectories with Inverse Kinematics and Observing the Robot's Response

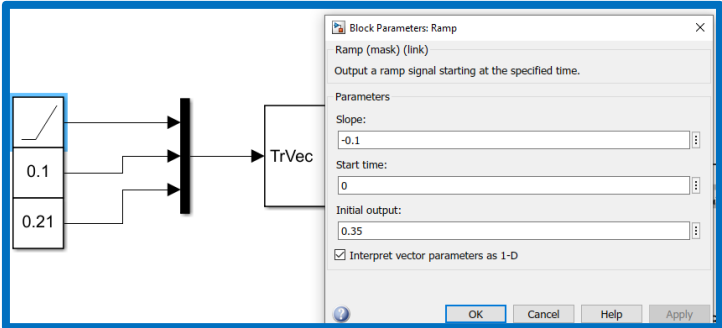
Step	Action	Screenshot
1	A First, we need to insert a MUX before the Coordinate Transformation Conversion block to be able to input XYZ translation vector coordinates. Select the Mux and make the number of ports >> 3	
	B - Insert a Constant block - Set it to 0.21 - Connect it to the 3rd port of the MUX (the Z port)	

Scenario A Horizontal Straight line		
Step	Action	Screenshot
2	A - Insert a Constant block - Set it to 0.10 - Connect it to the 2nd port of the MUX (the Y port)	
	B - Insert a Ramp block Ramp Block: - Slope >> - 0.1 - Start time >> 0 - Initial output >> 0.30 - Connect it to the 1st port of the MUX (the X port)	

Observing the Results:

Smooth horizontal motion with slight acceleration in the middle

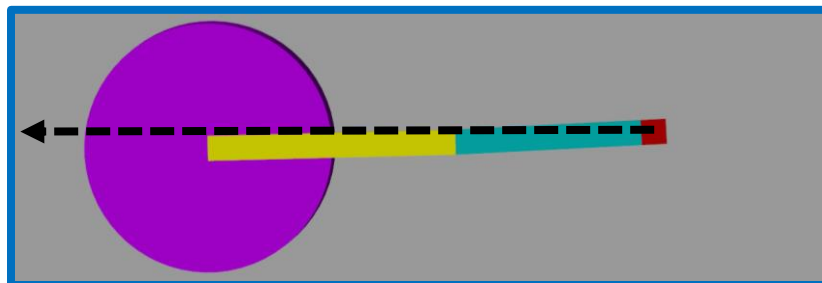


Scenario B Horizontal Straight line		
Step	Action	Screenshot
2	A - Insert a Constant block - Set it to 0.02 - Connect it to the 2nd port of the MUX (the Y port)	
	B - Insert a Ramp block Ramp Block: - Slope >> - 0.1 - Start time >> 0 - Initial output >> 0.35 - Connect it to the 1st port of the MUX (the X port)	

Observing the Results:

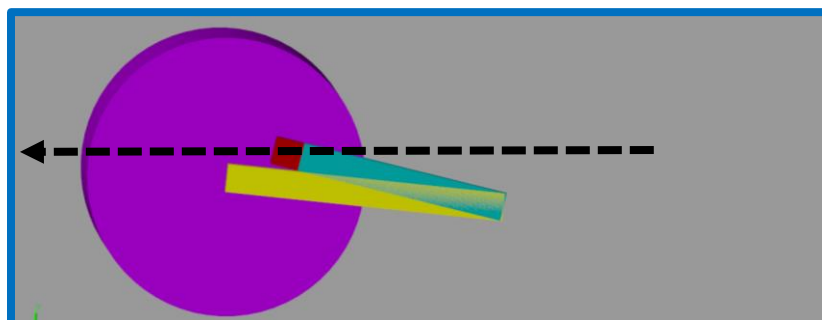
Very fast acceleration and speed at the beginning

(when near fully extended elbow position) ($q_2=0$)



Very fast acceleration and speed at the middle

(when near fully closed elbow position) ($q_2=180$)



Summary

